

Lab 01: Two Dimensional Arrays

Objective

In this lab, you will learn:

- Declaring and initializing Two-Dimensional Array.
- Processing Two-Dimensional Arrays.
- Passing Two-Dimensional Arrays to Methods

Practice Description: Grading Multiple Choice Test

Write a program that grades multiple-choice test. Suppose there are 4 students and 6 questions, and the answers are stored in a two dimensional array. Each row records a student's answers to the questions, as shown in the following array:

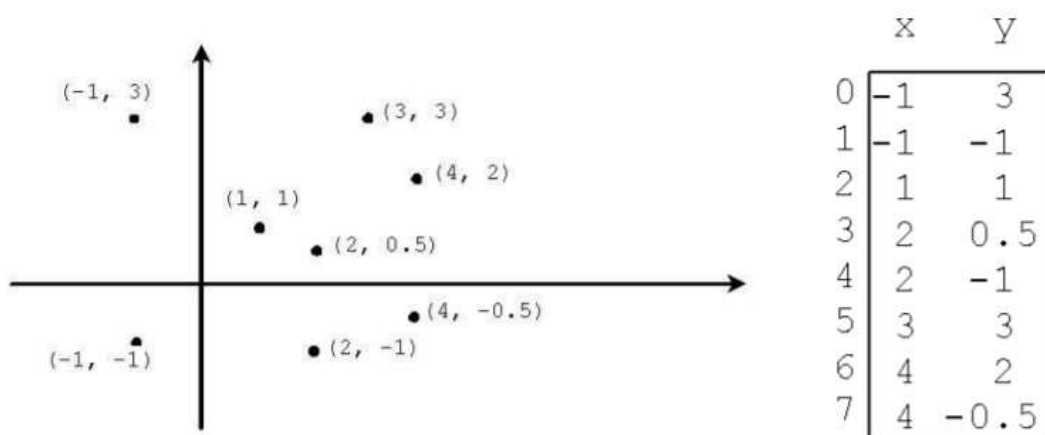
Students Answers to the Questions							Key Answers to Questions						
	0	1	2	3	4	5		0	1	2	3	4	5
Student 0	A	B	A	C	C	D	Key	D	B	D	C	C	D
Student 1	D	B	A	B	C	A							
Student 2	E	D	D	A	C	B							
Student 3	C	B	A	E	D	C							

Sample Output:

Student 0's correct count is 4 out of 6
Student 1's correct count is 3 out of 6
Student 2's correct count is 2 out of 6
Student 3's correct count is 1 out of 6

Activity Description: Finding a Closest Pair

The GPS navigation system is becoming increasingly popular. The system uses the graph and geometric algorithms to calculate distances and map a route. The section presents a geometric problem for finding a closest pair of point where points can be represented in two dimensional array.



Given a set of points, the closest-pair problem is to find the two points that are nearest to each other.
Distance between any two points (x_1, y_1) , (x_2, y_2) for example is as follow:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Notes:

You have to write 2 methods:

1. After filling the two dimensional array with the available points, send it to “**FindClosestTwoPoints**” method which will find and print the result
2. Since you need to find the distance between each 2 points, it is preferable to write a method “**Distance**” specifically for that.

Sample Output:

```
Enter the number of points: 8
Enter 8 points: -1 3 -1 -1 1 1 2 0.5 2 -1 3 3 4 2 4 -0.5
The closest two points are (1.0, 1.0) and (2.0, 0.5)
```